

WASP-104b

R-2565

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_170224 · created 2026-08-26T23:28:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457808.68208 +/- 0.00074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.037
Transit depth (Rp/Rs)^2	2.24 +/- 1.11 [%]
Orbital Inclination (inc)	82.38 +/- 2.1
Airmass coefficient 1 (a1)	0.988 +/- 0.02
Airmass coefficient 2 (a2)	0.0091 +/- 0.009
Scatter in the residuals of the lightcurve fit is	2.06 %
Best Comparison Star	#3 - [315, 417]
Optimal Method	PSF photometry
Transit Duration (day)	0.057 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.15 ± 0.037 vs 0.1214 (deviation 0.77σ)
Duration fit vs published	0.057 d vs 0.0742 d (deviation 0.69σ)
Mid-time O–C	1.5 min
Depth SNR	4.1
Residual scatter	2.06 %

Light curve

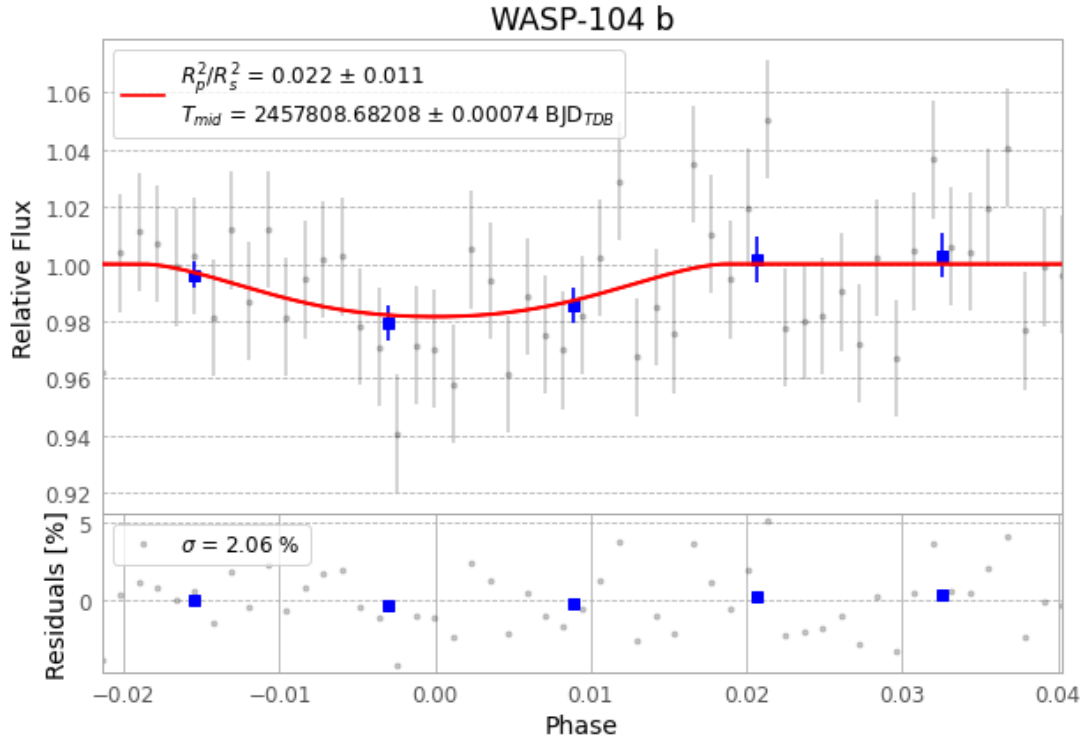


Figure 1 — FinalLightCurve_WASP-104 b_2017-02-23.png (SHA-256 17afe7b2c0bbbfbd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [537, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 825d85b74e8a346c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

53 FITS frames (35 MB), WASP-104170224031813.FITS → WASP-104170224055412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-170224.log (090ad4509452...), FinalLightCurve_WASP-104 b_2017-02-23.png (17afe7b2c0bb...), FinalLightCurve_WASP-104 b_2017-02-23.csv (5fc1b277e626...)

Compute resources

Wall time 92.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:28Z.